

OptimalControl.jl: A Julia package for modeling and solving optimal control problems with ODEs

Jean-Baptiste Caillau ¹, Olivier Cots ^{2¶}, Joseph Gergaud ², Pierre Martinon ³, and Sophia Sed⁴

¹ Université Côte d'Azur, CNRS, Inria, LJAD, France ² Université Toulouse, CNRS, ENSEEIHT-IRIT, France ³ CAGE team, Inria Paris, France ⁴ Inria Sophia Antipolis Méditerranée, France ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

[OptimalControl.jl](#) (Caillau, Cots, Gergaud, Martinon, & Sed, 2024) is a Julia (Bezanson et al., 2017) package for modeling and solving optimal control problems governed by ordinary differential equations (ODEs). As the core of the [control-toolbox ecosystem](#), it provides a unified framework that supports both direct and indirect solution methods with applications spanning aerospace, medical imaging, epidemiology, and quantum control.

The package features an expressive domain-specific language (DSL) built around the `@def` macro, enabling users to define control problems using notation that closely resembles standard mathematical formulations. Problems are solved through direct transcription, converting the continuous problem into a nonlinear program (NLP) using discretization schemes including Euler, trapezoidal, midpoint, and high-order Gauss-Legendre collocation. Alternatively, indirect shooting methods based on Pontryagin's Maximum Principle can be employed. The architecture relies on a modeler-solver separation that provides a modular and extensible foundation, enabling GPU execution (currently on NVIDIA hardware via [ExaModels.jl](#) and [MadNLP.jl](#)) with minimal user intervention alongside standard CPU solvers.

The ecosystem includes extensive [tutorial resources](#), a benchmark problem collection ([OptimalControlProblems.jl](#) (Caillau, Cots, Gergaud, & Martinon, 2024) with formulations in OptimalControl DSL and [JuMP](#) (Lubin et al., 2023)), and performance comparison tools ([CTBenchmarks.jl](#)). Integration with Julia's ecosystem enables access to state-of-the-art tools: NLP solvers [IPOPT](#) (Wächter & Biegler, 2006) and [MadNLP.jl](#) (Shin et al., 2021, 2024), automatic differentiation and NLP modeling through JuliaSmoothOptimizers' [ADNLPModels](#) (Migot et al., 2023), GPU acceleration via [ExaModels.jl](#) (Shin et al., 2023), numerical integration from SciML's [DifferentialEquations.jl](#) (Rackauckas & Nie, 2017), and visualization through [Plots.jl](#) (Christ et al., 2023).

Statement of Need

[OptimalControl.jl](#) is, to our knowledge, the only Julia package that unifies both direct and indirect methods for optimal control within a single, coherent framework.

This unified approach addresses a gap in the current optimal control software landscape. Existing tools are fragmented across programming languages and paradigms. Legacy packages such as [HamPath](#) (Caillau et al., 2012), [NutoPy](#) (Caillau et al., 2022), and [COTCOT](#) (Bonnard et al., 2007) implement sophisticated indirect methods but rely on Fortran implementations with MATLAB or Python interfaces, leading to complex and less extensible workflows. Proprietary solvers like [GPOPS-II](#) (Patterson & Rao, 2014) limit transparency and reproducibility. Open-

source tools such as [BOCOP](#) (Bonnans et al., 2017), [ACADO](#) (Houska et al., 2011), and [acados](#) (Verschuere et al., 2021) provide valuable direct method implementations but are not natively integrated into a modern, high-level scientific computing ecosystem. Python-based tools such as [Dymos](#) (Falck et al., 2021), [Pyomo.DAE](#) (Nicholson et al., 2018), [do-mpc](#) (Fiedler et al., 2023), and [GEKKO](#) (Beal et al., 2018) offer complementary capabilities but are similarly limited to direct transcription approaches within the Python ecosystem.

Within Julia, most existing packages target specific domains: [RobustAndOptimalControl.jl](#) for linear systems, [QuantumControl.jl](#) and [Piccolo.jl](#) for quantum optimal control, [DirectTrajectoryOptimization.jl](#) for trajectory problems, [LinearMPC.jl](#) (Arnström et al., 2022) and [ModelPredictiveControl.jl](#) (Gagnon et al., 2024) for model predictive control, and [InfiniteOpt.jl](#) (Pulsipher et al., 2022) for infinite-dimensional optimization. In contrast with [InfiniteOpt.jl](#) that is designed as an extension of [JuMP.jl](#), our package provides a modeler-agnostic approach, accepting general Julia code and leveraging various optimization solvers. Besides, contributing to general-purpose NLP modeling frameworks like [JuMP](#) or [CasADi](#) (Andersson et al., 2019) would not address the specific needs of optimal control: these tools lack native support for shooting methods based on Pontryagin's Maximum Principle and that heavily rely on differential geometric primitives.

[OptimalControl.jl](#) fills this gap by providing a DSL that matches mathematical notation, multiple discretization schemes and shooting methods, with planned support for homotopy continuation methods. The modeler-solver separation enables complementary use with [InfiniteOpt.jl](#) (JuMP-based) through alternative NLP modeling backends (currently [ADNLPModels](#), [ExaModels](#)) and solvers. GPU acceleration and an ecosystem with domain-specific applications, tutorials, and benchmarking tools complete the offering. Target users include researchers and engineers working in optimal control, control theorists developing new algorithms, and students learning optimal control through interactive tutorials.

State of the Field

[OptimalControl.jl](#) requires Julia version 1.10 or later and is registered in the Julia General registry, enabling straightforward installation via `Pkg.add("OptimalControl")`. We compare it below with existing software.

- **Legacy tools (COTCOT, HamPath, NutoPy):** These Fortran packages excel at indirect methods and homotopy continuation but require multi-language setup (Fortran plus MATLAB / Python). [OptimalControl.jl](#) provides both direct and indirect methods in pure Julia with straightforward installation via package manager.
- **Direct method tools (BOCOP, ACADO, GPOPS-II, acados, nosnoc):** Strong direct method implementations: [GPOPS-II](#) (Patterson & Rao, 2014) delivers mature methods with MATLAB and C++ implementations but is proprietary; [acados](#) (Verschuere et al., 2021) targets real-time MPC on embedded systems; [nosnoc](#) (Nurkanović & Diehl, 2022) specializes in nonsmooth optimal control; [CasADi](#) (Andersson et al., 2019), used as symbolic backend by several of these tools, is a general NLP modeler rather than an optimal control solver. [OptimalControl.jl](#) offers an open-source alternative with expressive DSL, native Julia ecosystem integration, GPU support, and unified direct and indirect approaches.
- **Julia packages:** [RobustAndOptimalControl.jl](#) targets linear systems; [QuantumControl.jl](#), [Piccolo.jl](#) and [DirectTrajectoryOptimization.jl](#) serve specific domains; [LinearMPC.jl](#) (Arnström et al., 2022) and [ModelPredictiveControl.jl](#) (Gagnon et al., 2024) focus on model predictive control. [InfiniteOpt.jl](#) addresses a very rich range of problems, including optimization on PDE's or with chance constraints, focusing on direct transcription methods. [OptimalControl.jl](#) supports both CPU and GPU execution (currently NVIDIA-only, via [ExaModels.jl](#) + [MadNLP.jl](#)), and adds tools to do shooting in a unified framework plus systematic benchmarking through [OptimalControlProblems.jl](#) and [CTBenchmarks.jl](#).

Illustrative Example

A self-contained illustrative example is provided as a companion repository (Caillau et al., 2026), archived on Zenodo. It combines direct and indirect solution approaches for a constrained energy minimization problem: a direct method on a coarse grid identifies the three-arc structure (unconstrained–constrained–unconstrained) and initializes a shooting method based on Pontryagin’s Maximum Principle, which then converges to arbitrary precision.

Software Design

The package architecture balances expressiveness, performance, and extensibility through modular design. The core is organized across internal packages within the [control-toolbox organization](#), each hosted as an independent Julia package and registered in the General registry: [CTBase.jl](#) defines the core types (optimal control models, solutions, initial guesses) and their accessors; [CTParser.jl](#) implements the `@def` DSL macro; [CTDirect.jl](#) handles direct transcription and NLP interfacing; and [CTFlows.jl](#) provides Hamiltonian flows and shooting for indirect methods. [OptimalControl.jl](#) re-exports these packages as a unified entry point. Contributors interested in a specific functionality can work directly on the relevant sub-package, each of which has its own documentation, tests, and continuous integration. The delegation of specific responsibilities is as follows:

- **DSL parsing:** [MLStyle.jl](#) (Thautwarm & contributors, 2023) enables pattern matching on abstract syntax trees generated from `@def` macro invocations. This design choice separates problem specification from solution methods, and allows a syntax as close as possible to the mathematical description of the problem.
- **Problem models and solutions:** Structured types represent optimal control models, solutions, and initial guesses. Each type provides textual visualization, accessor methods for introspection, and visual plotting capabilities for solutions. Extensible abstractions enable addition of new problem classes without modifying core code.
- **Discretization:** Direct transcription converts continuous problems into NLPs. Supporting multiple discretization schemes (Euler, midpoint, trapezoidal, Gauss-Legendre) required careful abstraction to share transcription logic while allowing scheme-specific implementations.
- **Modelers and solvers:** We chose a clear modeler-solver separation: discretization produces [ADNLPModels](#) (able to deal with arbitrary user defined functions) or [ExaModels](#) (Shin et al., 2023) instances compatible with multiple NLP solvers ([IPOPT](#) via [NLPModelsIpopt.jl](#) (Orban et al., 2023), [Knitro](#) (Byrd et al., 2006), [MadNLP](#), and [Uno](#) (Vanaret & Leyffer, 2026)). These NLP solvers rely on linear solvers such as [MUMPS.jl](#) (Amestoy et al., 2001, 2019; Montoison et al., 2024) for CPU computations or [CUDSS.jl](#) (NVIDIA Corporation, 2025) for GPU acceleration with [MadNLP](#). Automatic differentiation via [ForwardDiff.jl](#) (Revels et al., 2016) and [DifferentiationInterface.jl](#) (Dalle & Hill, 2026) avoids manual derivative coding.
- **Indirect methods:** Hamiltonian flows integrate with [DifferentialEquations.jl](#) to access adaptive stepping, event handling, and multiple ODE solvers without reimplementing. Shooting methods rely on [NonlinearSolve.jl](#) (Pal et al., 2025) or [MINPACK.jl](#) for root-finding. Future extensions will incorporate homotopy continuation methods leveraging bifurcation analysis tools like [BifurcationKit.jl](#) (Veltz, 2020).
- **Testing and quality assurance:** The multi-repository structure is matched by a layered testing strategy. Each sub-package is tested independently, combining unit tests, integration tests, and code-quality checks, while [OptimalControl.jl](#) adds strong end-to-end integration tests that solve complete optimal control problems by both direct and indirect methods. Continuous integration runs across Linux, macOS, and Windows, on

both CPU and GPU (through a self-hosted CUDA runner), using reusable GitHub Actions workflows centralized in [CTActions](#) and shared across the ecosystem. Code coverage is tracked on Codecov, downstream packages and applications are guarded against regressions through dedicated breakage tests triggered on pull requests, and beta versions are distributed during development via a local registry, [ct-registry](#). Non-regression on complex problems is increasingly driven by [OptimalControlProblems.jl](#), used as a curated problem bank.

Key design trade-offs:

1. **DSL vs. programmatic API:** We prioritized a mathematical DSL to reduce cognitive load for domain experts, accepting increased parsing complexity handled internally. Currently we do not address optimization of PDE or stochastic systems.
2. **GPU acceleration strategy:** The modeler-solver separation enables a straightforward CPU-to-GPU transition. Users select the ExaModels modeler with MadNLP and the CUDSS linear solver for GPU execution, or simply append the `:gpu` token to the `solve` call. This modular approach minimizes maintenance burden while enabling GPU performance without reimplementing transcription logic. GPU support is currently limited to NVIDIA hardware and to the ExaModels + MadNLP modeler-solver combination.
3. **Method coverage:** Supporting both direct and indirect approaches increases code complexity but serves distinct user needs: direct methods for constrained problems with many variables, indirect methods for theoretical analysis and smaller problems requiring high accuracy. Both approaches resort to iterative solvers and may converge or not, *e.g.* depending on the initial guess. In the case of optimization solvers, the full output status of the solver is returned allowing *a posteriori* analysis.

Research Impact Statement

OptimalControl.jl has been applied in published research across multiple domains. Independent external adoption demonstrates the package's broader impact: Ferde et al. (2025) used it for drone racing trajectory optimization, Caio & Silva (2025) for epidemiological models using indirect methods, Morsky (2025) for vaccination strategies under social norms, Opmeer (2025) for optimal harvesting of age-structured populations, and Evangelakos et al. (2025) for fast charging protocols for quantum batteries.

Research by close collaborators further demonstrates the package's versatility across aerospace (Herasimenka et al. (2026)), epidemiology (Bliman et al. (2026)), quantum control (Beschastnyi et al. (2025)), microbial growth control (Innerarity Imizcoz et al. (2025)), and theoretical optimal control (Bouali et al. (2025); Lutz & Privat (2025); Bayen et al. (2026); Bonnard et al. (2026)).

The [control-toolbox organization](#) hosts domain-specific application packages built on OptimalControl.jl: medical imaging optimization ([MagneticResonanceImaging.jl](#)), gene regulatory networks ([PWLdynamics.jl](#)), spacecraft orbital transfers ([Kepler.jl](#)), epidemiological modeling ([SIRcontrol.jl](#)), and variational calculus ([CalculusOfVariations.jl](#)).

The package serves educational purposes through [Tutorials.jl](#) covering topics from linear-quadratic regulators to Model Predictive Control. These resources are used in academic courses and workshops; the package was notably presented at JuliaCon 2023 (Caillaud et al., 2023), with the documentation serving as the presentation support. [OptimalControlProblems.jl](#) provides standardized test problems formulated in both OptimalControl DSL and JuMP, enabling systematic performance comparisons through [CTBenchmarks.jl](#).

Development involves international collaborations with CNES (French space agency), Thales Alenia Space, Inria, and CNRS. The package development began in September 2022, with the first public release in February 2023. As of March 2026, OptimalControl.jl has published

187 more than 40 releases with contributions from multiple developers across the control-toolbox
188 ecosystem packages, demonstrating sustained community engagement.

189 AI Usage Disclosure

190 The core software implementation, algorithms, and architectural design of OptimalControl.jl
191 were developed by the authors. Generative AI tools—specifically Claude Sonnet (version 4.6)
192 and Claude Opus (version 4.8) by Anthropic—have been used to assist with code refactoring
193 and generation of unit and integration tests, always under human review. This paper was
194 drafted by the authors and subsequently revised with AI assistance for restructuring according
195 to JOSS format requirements and language editing. All technical content, examples, and
196 claims reflect the authors' work and judgment.

197 Acknowledgements

198 We thank the [control-toolbox community](#) and the broader Julia ecosystem for contributions
199 that shaped the package design and educational resources. Development has been supported
200 by partnerships with CNES, Thales Alenia Space, Inria, and CNRS. The software is distributed
201 under the MIT license (see [LICENSE](#)). We welcome community contributions following the
202 guidelines in [CONTRIBUTING.md](#).

203 References

- 204 Amestoy, P. R., Buttari, A., L'Excellent, J.-Y., & Mary, T. (2019). Performance and Scalability
205 of the Block Low-Rank Multifrontal Factorization on Multicore Architectures. *ACM*
206 *Transactions on Mathematical Software*, 45(1), 2:1–2:26. <https://doi.org/10.1145/3242094>
- 207 Amestoy, P. R., Duff, I. S., Koster, J., & L'Excellent, J.-Y. (2001). A fully asynchronous
208 multifrontal solver using distributed dynamic scheduling. *SIAM Journal on Matrix Analysis*
209 *and Applications*, 23(1), 15–41. <https://doi.org/10.1137/s0895479899358194>
- 210 Andersson, J. A., Gillis, J., Horn, G., Rawlings, J. B., & Diehl, M. (2019). CasADi: A software
211 framework for nonlinear optimization and optimal control. *Mathematical Programming*
212 *Computation*, 11(1), 1–36. <https://doi.org/10.1007/s12532-018-0139-4>
- 213 Arnström, D., Bemporad, A., & Axehill, D. (2022). A dual active-set solver for embedded
214 quadratic programming using recursive LDL^T updates. *IEEE Transactions on Automatic*
215 *Control*, 67(8), 4362–4369. <https://doi.org/10.1109/TAC.2022.3176430>
- 216 Bayen, T., Bouali, A., & Bourdin, L. (2026). Minimum time problem for the double integrator
217 with a loss-of-control region. *Nonlinear Analysis: Hybrid Systems*, 60, 101681. <https://doi.org/10.1016/j.nahs.2026.101681>
- 218
- 219 Beal, L. D. R., Hill, D. C., Martin, R. A., & Hedengren, J. D. (2018). GEKKO optimization
220 suite. *Processes*, 6(8), 106. <https://doi.org/10.3390/pr6080106>
- 221 Beschastnyi, I., Dell'Elce, L., Pomet, J.-B., Sacchelli, L., & Tinoco, D. (2025). Pulse control of
222 affine systems with applications to quantum control. *2025 IEEE 64th Conference on Decision*
223 *and Control (CDC)*, 3769–3774. <https://doi.org/10.1109/CDC57313.2025.11312971>
- 224 Bezanson, J., Edelman, A., Karpinski, S., & Shah, V. B. (2017). Julia: A fresh approach to
225 numerical computing. *SIAM Review*, 59(1), 65–98. <https://doi.org/10.1137/141000671>
- 226 Bliman, P.-A., Bouali, A., Loisel, P., Rapaport, A., & Virelizier, A. (2026). On the problem
227 of minimizing the epidemic final size for SIR model by social distancing. *Mathematical*
228 *Biosciences and Engineering*, 23(3), 567–593. <https://doi.org/10.3934/mbe.2026022>

- 229 Bonnans, J. F., Martinon, P., Giorgi, D., Grélaud, V., Maindrault, S., & Tissot, O. (2017).
 230 The bocop and BocHop optimization toolboxes. *IFAC-PapersOnLine*, 50(1), 8849–8854.
 231 <https://doi.org/10.1016/j.ifacol.2017.08.1745>
- 232 Bonnard, B., Caillaud, J.-B., & Trélat, E. (2007). Second order optimality conditions in the
 233 smooth case and applications in optimal control. *ESAIM: Control, Optimisation and*
 234 *Calculus of Variations*, 13(2), 207–236. <https://doi.org/10.1051/cocv:2007012>
- 235 Bonnard, B., Cots, O., & Privat, Y. (2026). The zermelo navigation problem on the 2-
 236 sphere of revolution: An optimal control perspective with applications to micromagnetism.
 237 *Nonlinearity*, 39(1), 015006. <https://doi.org/10.1088/1361-6544/ae2cd9>
- 238 Bouali, A., Rapaport, A., & Bayen, T. (2025). *Regularization of optimal control problems on*
 239 *stratified domains using additional controls*. <https://hal.inrae.fr/hal-04928858>
- 240 Byrd, R. H., Nocedal, J., & Waltz, R. A. (2006). Knitro: An integrated package for nonlinear
 241 optimization. In G. Di Pillo & M. Roma (Eds.), *Large-scale nonlinear optimization* (pp.
 242 35–59). Springer US. https://doi.org/10.1007/0-387-30065-1_4
- 243 Caillaud, J.-B., Cots, O., & Gergaud, J. (2012). Differential continuation for regular optimal
 244 control problems. *Optimization Methods and Software*, 27(2), 177–196. <https://doi.org/10.1080/10556788.2011.593625>
- 245
 246 Caillaud, J.-B., Cots, O., Gergaud, J., & Martinon, P. (2024). *OptimalControlProblems.jl:*
 247 *a collection of optimal control problems with ODEs in Julia*. <https://doi.org/10.5281/zenodo.17013180>
- 248
 249 Caillaud, J.-B., Cots, O., Gergaud, J., Martinon, P., & Sed, S. (2023). *OptimalControl.jl: A*
 250 *Julia package to model and solve optimal control problems*. JuliaCon 2023, Cambridge, MA.
 251 Talk and abstract: <https://pretalx.com/juliacon2023/talk/HDC8F7/>. Video recording:
 252 <https://www.youtube.com/watch?v=RYUtVnzLj5k>.
- 253 Caillaud, J.-B., Cots, O., Gergaud, J., Martinon, P., & Sed, S. (2024). *OptimalControl.jl: a*
 254 *Julia package to model and solve optimal control problems with ODEs*. <https://doi.org/10.5281/zenodo.13336563>
- 255
 256 Caillaud, J.-B., Cots, O., Gergaud, J., Martinon, P., & Sed, S. (2026). *Joss-oc-example:*
 257 *Illustrative example companion to the OptimalControl.jl JOSS paper* (Version v1.0.0).
 258 Zenodo. <https://doi.org/10.5281/zenodo.19663650>
- 259
 260 Caillaud, J.-B., Cots, O., & Martinon, P. (2022). Ct: Control toolbox – numerical tools and
 261 examples in optimal control. *IFAC-PapersOnLine*, 55(16), 13–18. <https://doi.org/10.1016/j.ifacol.2022.08.074>
- 262
 263 Caio, P., & Silva, C. J. (2025). Application of indirect methods to optimal control problems in
 264 epidemiology. *CONTROLO 2024*, 444–454. https://doi.org/10.1007/978-3-031-81724-3_40
- 265
 266 Christ, S., Schwabeneder, D., Rackauckas, C., Borregaard, M. K., & Breloff, T. (2023).
 267 *Plots.jl – a user extendable plotting API for the julia programming language*. <https://doi.org/10.5334/jors.431>
- 268
 269 Dalle, G., & Hill, A. (2026). A common interface for automatic differentiation. *Journal of*
 270 *Machine Learning Research*, 27(25), 1–13. <http://jmlr.org/papers/v27/25-1024.html>
- 271
 272 Evangelakos, V., Paspalakis, E., & Stefanatos, D. (2025). Fast protocols for charging a
 273 three-spin-chain quantum battery. *Scientific Reports*, 15(1), 45626. <https://doi.org/10.1038/s41598-025-30354-1>
- 274
 275 Falck, R. D., Gray, J. S., Ponnappalli, K., & Wright, T. (2021). Dymos: A python package
 for optimal control of multidisciplinary systems. *Journal of Open Source Software*, 6(59),
 2809. <https://doi.org/10.21105/joss.02809>

- 276 Ferede, R., Blaha, T., Lucassen, E., De Wagter, C., & Croon, G. C. H. E. de. (2025). One net
277 to rule them all: Domain randomization in quadcopter racing across different platforms.
278 *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 6357–6363.
279 <https://doi.org/10.1109/ICRA55743.2025.11128790>
- 280 Fiedler, F., Karg, B., Lüken, L., Brandner, D., Heinlein, M., Brabender, F., & Lucia, S. (2023).
281 Do-mpc: Towards FAIR nonlinear and robust model predictive control. *Control Engineering*
282 *Practice*, 140, 105676. <https://doi.org/10.1016/j.conengprac.2023.105676>
- 283 Gagnon, F., Thivierge, A., Desbiens, A., & Bagge Carlson, F. (2024). *ModelPredictiveControl.jl:*
284 *advanced process control made easy in Julia*. <https://doi.org/10.48550/arXiv.2411.09764>
- 285 Herasimenka, A., Baresi, N., Green, L., Morgan, H., Underwood, C. I., Bridges, C., Jason,
286 S., Lucca Fabris, A., & Ryden, K. (2026). Low-thrust trajectory optimization for the
287 moon-enabled sun occultation mission concept. *AIAA SCITECH 2026 Forum*.
- 288 Houska, B., Ferreau, H. J., & Diehl, M. (2011). ACADO toolkit—an open-source framework for
289 automatic control and dynamic optimization. *Optimal Control Applications and Methods*,
290 32(3), 298–312. <https://doi.org/10.1002/oca.939>
- 291 Innerarity Imizcoz, J., Yabo, A. G., Djema, W., Francis, F., & Gouzé, J.-L. (2025). *Optimal*
292 *allocation control in microbial growth under a heat-shock*. [https://inria.hal.science/hal-](https://inria.hal.science/hal-05369609)
293 [05369609](https://inria.hal.science/hal-05369609)
- 294 Lubin, M., Dowson, O., Dias Garcia, J., Huchette, J., Legat, B., & Vielma, J. P. (2023).
295 JuMP 1.0: Recent improvements to a modeling language for mathematical optimization.
296 *Mathematical Programming Computation*. <https://doi.org/10.1007/s12532-023-00239-3>
- 297 Lutz, K., & Privat, Y. (2025). *Minimal time control of underdamped parametric oscillators*.
298 <https://hal.science/hal-05047678>
- 299 Migot, T., Montoison, A., Orban, D., Soares Siqueira, A., & contributors. (2023).
300 *ADNLPMODELS.jl: Automatic Differentiation models implementing the NLPModels API*.
301 <https://github.com/JuliaSmoothOptimizers/ADNLPMODELS.jl>
- 302 Montoison, A., Orban, D., Sweeney, W. R., & contributors. (2024). *MUMPS.jl: A*
303 *Julia Interface to MUMPS* (Version 1.4.1). [https://github.com/JuliaSmoothOptimizers/](https://github.com/JuliaSmoothOptimizers/MUMPS.jl)
304 [MUMPS.jl](https://github.com/JuliaSmoothOptimizers/MUMPS.jl)
- 305 Morsky, B. (2025). Vaccination and collective action under social norms. *Bulletin of*
306 *Mathematical Biology*, 87(5), 55. <https://doi.org/10.1007/s11538-025-01436-y>
- 307 Nicholson, B., Sirola, J. D., Watson, J.-P., Zavala, V. M., & Biegler, L. T. (2018). Pyomo.dae:
308 A modeling and automatic discretization framework for optimization with differential
309 and algebraic equations. *Mathematical Programming Computation*, 10(2), 187–223.
310 <https://doi.org/10.1007/s12532-017-0127-0>
- 311 Nurkanović, A., & Diehl, M. (2022). Nosnoc: A software package for numerical optimal
312 control of nonsmooth systems. *IEEE Control Systems Letters*, 6, 3110–3115. <https://doi.org/10.1109/lcsys.2022.3181800>
- 313
- 314 NVIDIA Corporation. (2025). *NVIDIA cuDSS (Preview): A high-performance CUDA Library*
315 *for Direct Sparse Solvers*. <https://docs.nvidia.com/cuda/cudss/index.html>.
- 316 Opmeer, M. (2025). Optimal harvesting of a continuously age-structured population with
317 density dependence. *PLOS ONE*, 20(9), 1–12. [https://doi.org/10.1371/journal.pone.](https://doi.org/10.1371/journal.pone.0333087)
318 [0333087](https://doi.org/10.1371/journal.pone.0333087)
- 319 Orban, D., Soares Siqueira, A., & contributors. (2023). *NLPModelsIpopt.jl: A thin IPOPT*
320 *wrapper for NLPModels* (Version 0.10.1). [https://github.com/JuliaSmoothOptimizers/](https://github.com/JuliaSmoothOptimizers/NLPModelsIpopt.jl)
321 [NLPModelsIpopt.jl](https://github.com/JuliaSmoothOptimizers/NLPModelsIpopt.jl)
- 322 Pal, A., Holtorf, F., Larsson, A., Loman, T., Utkarsh, Schäfer, F., Qu, Q., Edelman, A., &

- 323 Rackauckas, C. (2025). NonlinearSolve.jl: High-performance and robust solvers for systems
324 of nonlinear equations in julia. *ACM Trans. Math. Softw.* [https://doi.org/10.1145/](https://doi.org/10.1145/3779117)
325 [3779117](https://doi.org/10.1145/3779117)
- 326 Patterson, M. A., & Rao, A. V. (2014). GPOPS-II: A MATLAB software for solving multiple-
327 phase optimal control problems using hp-adaptive gaussian quadrature collocation methods
328 and sparse nonlinear programming. *ACM Transactions on Mathematical Software*, 41(1),
329 1–37. <https://doi.org/10.1145/2558904>
- 330 Pulsipher, J. L., Zhang, W., Hongisto, T. J., & Zavala, V. M. (2022). A unifying modeling
331 abstraction for infinite-dimensional optimization. *Computers & Chemical Engineering*, 156.
332 <https://doi.org/10.1016/j.compchemeng.2021.107567>
- 333 Rackauckas, C., & Nie, Q. (2017). Differentialequations.jl—a performant and feature-rich
334 ecosystem for solving differential equations in julia. *Journal of Open Research Software*,
335 5(1), 15.
- 336 Revels, J., Lubin, M., & Papamarkou, T. (2016). *Forward-mode automatic differentiation in*
337 *Julia*. <https://arxiv.org/abs/1607.07892>
- 338 Shin, S., Anitescu, M., & Pacaud, F. (2024). Accelerating optimal power flow with GPUs: SIMD
339 abstraction of nonlinear programs and condensed-space interior-point methods. *Electric*
340 *Power Systems Research*, 236, 110651. <https://doi.org/10.1016/j.epsr.2024.110651>
- 341 Shin, S., Coffrin, C., Sundar, K., & Zavala, V. M. (2021). Graph-based modeling and
342 decomposition of energy infrastructures. *IFAC-PapersOnLine*, 54(3), 693–698. <https://doi.org/10.1016/j.ifacol.2021.08.322>
- 343 Shin, S., Pacaud, F., & Anitescu, M. (2023). *Accelerating optimal power flow with GPUs:*
344 *SIMD abstraction of nonlinear programs and condensed-space interior-point methods.*
345 <https://doi.org/10.2139/ssrn.4601442>
- 346 Thautwarm, & contributors. (2023). *MLStyle.jl: Functional programming and pattern*
347 *matching for Julia*. <https://thautwarm.github.io/MLStyle.jl>
- 348 Vanaret, C., & Leyffer, S. (2026). *Implementing a unified solver for nonlinearly constrained*
349 *optimization*.
- 350 Veltz, R. (2020). *BifurcationKit.jl*. Inria Sophia-Antipolis. [https://hal.archives-ouvertes.fr/hal-](https://hal.archives-ouvertes.fr/hal-02902346)
351 [02902346](https://hal.archives-ouvertes.fr/hal-02902346)
- 352 Verschueren, R., Frison, G., Kouzoupis, D., Frey, J., Duijkeren, N. van, Zanelli, A., Novoselnik,
353 B., Albin, T., Quirynen, R., & Diehl, M. (2021). Acados – a modular open-source
354 framework for fast embedded optimal control. *Mathematical Programming Computation*.
355 <https://doi.org/10.1007/s12532-021-00208-8>
- 356 Wächter, A., & Biegler, L. T. (2006). On the implementation of an interior-point filter
357 line-search algorithm for large-scale nonlinear programming. *Mathematical Programming*,
358 106(1), 25–57. <https://doi.org/10.1007/s10107-004-0559-y>
- 359